

Digital System Design

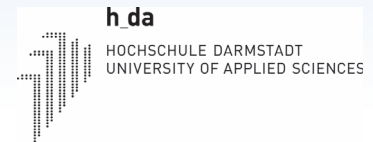
Lecture SS 2023

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OUTLINE

A Low Power Design

A1: Power dissipation in CMOS

A1.1: dynamic power, static power

A1.2: power challenge

A2: Design techniques

A2.1: FinFET

A2.2: Multi Vdd, Power Gating, Multi-Vt-voltage libraries

A3: Trends

A3.1: self-timed/asynchronous design

A3.2: wave pipelining

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OUTLINE

B High Performance Design

B1: Parallel Processing

B1.1: sample speed optimization

B1.2: low power

B2: Pipelining

B2.1: throughput optimization

B2.2: low power

B3: Reduction of Clock Skew

B3.1: clock skew impact

B3.2: clock distribution networks

B4: Trends

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OUTLINE

C Arithmetic Circuits

C1: Adder

C1.1: full adder design

C1.2: carry-ripple/carry-select/carry-lookahead adder

C2: Multiplier

C2.1: array multiplier with CPA

C2.2: array multiplier with CSA

D Hardware Description Language VHDL

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Textbook [1]:

N. Weste, D. Harris:
CMOS VLSI Design
A Circuits and Systems Perspective
Fourth Edition (US), 2010, Addison-Wesley
Integrated Circuit Design
Fourth Edition (Global Edition) 2011, Pearson Education

for some specific topics please refer to [2]:

J. Rabaey, A. Chandrakasan, B. Nikolic:
Digital Integrated Circuits
A Design Perspective
Second Edition, 2003, Prentice Hall

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Textbook [3] for design of digital signal processing systems:

Keshab K. Parhi:
VLSI Digital Signal Processing Systems: Design and
Implementation, John Wiley & Sons, 1999.